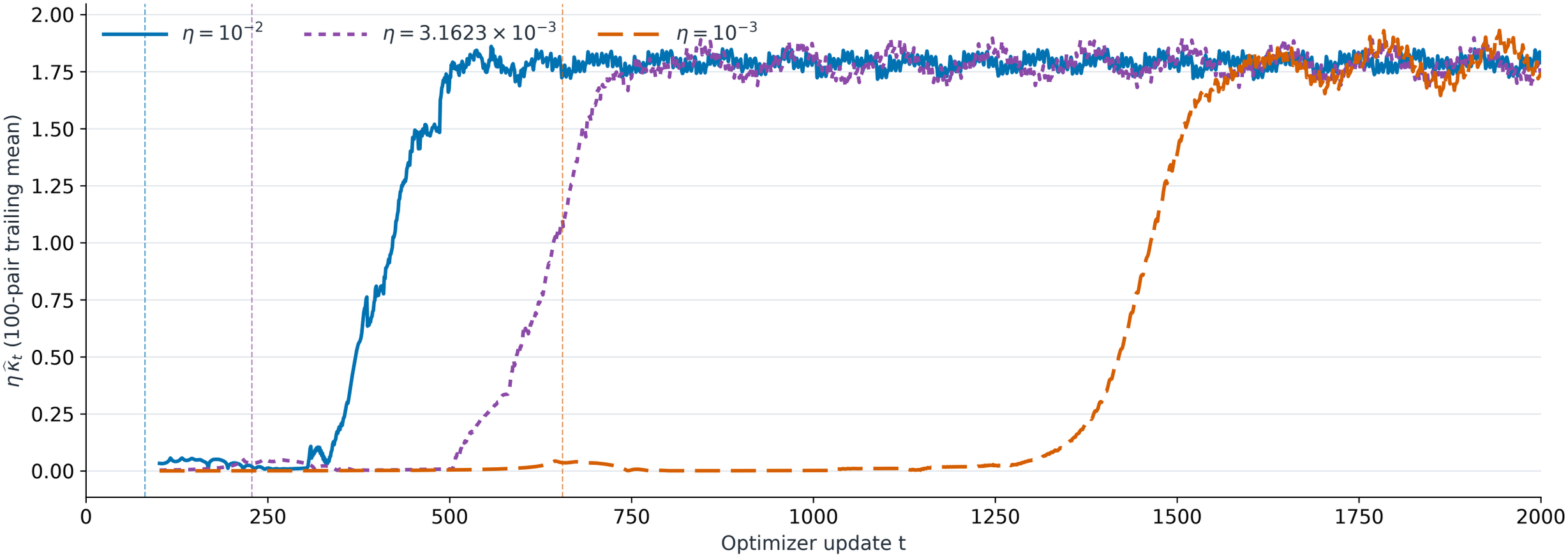
Rosenbrock: learning-rate-scaled gradient-identity kappa

Three matched AdamW learning rates | Start (0, 1) | 2,000 updates | Weight decay 0.1

$$\eta\,\widehat{\kappa}_t = 1 - \rho_t\,\|g_t\|/\|g_{t-1}\| \quad (\varepsilon_t = 0)$$



Multiply each unsmoothed kappa proxy by its constant learning rate, then average the last 100 finite adjacent pairs.

First complete mean at t = 101. All 1,900 means per LR are shown; dashed vertical lines mark the original first y-turns.

Dimensionless proxy; epsilon = 0 (exact gradients). The SGD-derived identity is evaluated along AdamW; no stability threshold is implied.